



Statistics of Financial Markets

Chapter 7: Stochastic Volatility Models – Part 1



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Outline

Motivation

ARCH Model

GARCH Model

Estimation

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References

Motivation

- Why constant volatility models are inadequate for financial data
- Volatility clustering and the need for conditional heteroskedasticity models
- Overview of the ARCH/GARCH modeling framework

ARCH Model

- ▣ Engle's (1982) ARCH(q) specification
- ▣ Conditional variance: $\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$
- ▣ Stationarity conditions and unconditional variance
- ▣ Properties: excess kurtosis and volatility clustering

GARCH Model

- Bollerslev's (1986) GARCH(p, q) generalization
- Conditional variance: $\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$
- Persistence and the IGARCH case
- GARCH(1,1) as the workhorse model

Estimation

- ▣ Maximum Likelihood Estimation (MLE)
- ▣ Log-likelihood function for GARCH models
- ▣ Quasi-MLE under non-Gaussian innovations
- ▣ Numerical optimization: BFGS, constraints

Model Selection

- Information criteria: AIC, BIC, HQIC
- Likelihood ratio tests for nested models
- Residual diagnostics: standardized residuals, Ljung–Box test on $\hat{z}_t^2$
- ARCH-LM test for remaining heteroskedasticity

Applications

- ▣ Fitting ARCH/GARCH models to stock index returns
- ▣ Volatility forecasting and comparison with realized volatility
- ▣ Python implementation using `arch` package

 SFM_ch7_garch

Conclusions

- ▣ Summary of the ARCH and GARCH modeling framework
- ▣ Strengths and limitations of symmetric volatility models
- ▣ Preview of asymmetric extensions in Chapter 8

References

- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity.
- Francq, C., Zakoian, J.-M. (2010). GARCH Models: Structure, Statistical Inference and Financial Applications.